MGT 446 Spreadsheet Modeling

Fall 2026

Prof. Robert Montgomery
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Office Hours: *in person* Thursdays 3:00pm-4:30pm (excluding 11/26)
Wednesdays 4:00pm-5:30pm (excluding 11/11 & 11/25)
or via Zoom by appointment

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CLASSROOM LOCATION

OTRSN 1E107
Wednesdays 6:30pm-9:20pm

DESCRIPTION

This course focuses on managerial modeling approaches used to make better business decisions. Students will be exposed to an array of models for optimization and decision making under uncertainty. The emphasis of the course will be on translating problems into mathematical models that can be solved using tools available in Microsoft Excel, and then analyzing and interpreting the results. This approach is meant to make the material approachable for all MBA students, and while we will at times discuss the underlying theory and algorithm, students will not be tested on this material. This course is designed around Microsoft Excel because of its widespread use and accessibility.

The course will be comprised of two units, the first will focus on deterministic models, where we will use programs to sort through many possible options to arrive at problem solutions that meet the desired criteria. Students will learn how to define objective functions and constraints in a way that leverages the abilities of computers to solve these types of problems and derive meaningful insights from sensitivity analysis. Following the Midterm, the second unit will focus on models with random variables. These models will build on the those from the first unit, while introducing other modeling approaches, namely Monte Carlo simulations. The course content is built on a wide range of practical cases from various industries for students to explore through hands-on collaborative work.

OBJECTIVES

At the close of this course, you will be able to:

- Recognize instances where models can help improve business operations and decision making
- Translate managerial decisions into mathematical models
- Create streamlined and effective Microsoft Excel models
- Incorporate random variable distributions into decision making under uncertainty
- Clearly communicate modeling approaches and results so that the models are transferable to other users

MATERIALS

Required:

- Microsoft Excel (at least 2010, but ideally later)
- Open Solver (will discuss installation in class)
- Cases will be provided via Canvas

SCHEDULE

Date	Due	Class Topic
Week 1 September 30		Building Better Excel Models
Week 2 October 7	Assignment 1	Linear Optimization
Week 3 October 14	Assignment 2	Sensitivity Analysis
Week 4 October 21	Assignment 3	Binary and Integer Optimization Models
Week 5 October 28	**Midterm**	Monte Carlo Simulation
Week 6 November 4	Assignment 4	Simulation with Multiple Random Variables
Week 7 November 11 (No class Veterans Day)		
Week 8 November 18	Assignment 5	Simulation with Correlated Random Variables
Week 9 November 25 (No class Thanksgiving Week)		
Week 10 December 2	Assignment 6	Review

ASSIGNMENTS

There will be a total of six assignments. All assignments will be completed in groups. The group sizes will depend on the number of students enrolled in the course. Each assignment will require submission of an Excel spreadsheet with the completed model as well a summary document. The summary document can be either a written document or a summary video. At least two of the submissions must be summary videos. More information about formatting for these summaries will be provided during the first class.

Exams

There will be both a midterm and final exam. The midterm will take 75 minutes and be held during the class time of week 5 (October 28). We will hold an abbreviated class after the midterm exam.
The final exam will take place on Wednesday December 9, from 6:30-9:30pm in OTRSN 1E107. Please contact the instructor as soon as possible if you will not be available for either of these sessions, as accommodations can be made with enough advanced notice.

Class Participation Attendance and Participation

Because of the collaborative nature of this class and the emphasis on in-class participation, it is important that you attend every class. Please let the instructor know at the start of the quarter if you have any planned absences.

GRADING

Assignments	Percentages
Groups Assignments	42%
Participation	8%
Midterm	20%
Final	30%
Total	**100%**

AI TOOLs
LLMs and generative AI tools are now widely available. In your careers you will likely have to figure out when these tools are helpful and when they are not. You are welcome to use these tools for this class, but must cite all instances of use beyond simple searches. I ask that you use these tools to enhance your learning and work, not to replace your learning.

ACADEMIC INTEGRITY
Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2

STUDENTS WITH DISABILITIES
A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or **osd@ucsd.edu.**